

Soomin Lee

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Summary

Data-driven bioprocess engineer and AI researcher with experience spanning deep learning-based fermentation optimization and GMP-compliant large-scale biologics manufacturing.

Education

Korea Advanced Institute of Science and Technology Daejeon, South Korea	<i>Feb 2025 - Present</i>
Master's in Chemical and Biomolecular Engineering	Expected graduation: Feb 2026
Georgia Institute of Technology Atlanta, GA GPA: 3.75	<i>August 2021 - May 2024</i>
Bachelor of Science in Biomedical Engineering	

Experience

Contributor of Google Summer of Code (GSoC) - collaborating with European Molecular Biology Laboratory's European Bioinformatics Institute (EMBL-EBI) *May 2026 - Present*
Engineering an automated metadata extraction pipeline that mines full-text genomics literature to recover ploidy, sex, and strain fields missing from standardized genome-assembly submissions, combining rule-based extraction with a hybrid BM25 + PubMedBERT retrieval ensemble unified via reciprocal rank fusion.

Master's student at Systems Biology and Medicine Lab, KAIST *Feb 2025 - Present*

- Multimodal Deep Learning for Condition-Aware Gene & BGC Expression Prediction

Developing a Transformer-based multimodal deep learning framework to predict condition-dependent microbial gene and biosynthetic gene cluster (BGC) expression by integrating DNA sequence features, enzyme domain embeddings (Transformer/GCN), regulatory elements, host EC profiles, and media composition. Constructing large-scale RNA-seq-derived regulation datasets with imbalance-aware optimization and investigating media- and host-dependent BGC activation patterns for natural product discovery and precision metabolic engineering.

- Predictive Machine Learning Model for Optimized 2,3-Butanediol Fermentation

Developed BiLSTM and Transformer-based time-series forecasting models in collaboration with GS Caltex for industrial 2,3-BDO fermentation, integrating 250-strain genotype embeddings, medium composition vectors, and dynamic process parameters across 1,175 batches and 664k data points, achieving 3.14% relative MAE for BDO prediction. Built a real-time, forecast-driven control model that simulates constrained RPM adjustments within operational limits and proactively recommends optimal setpoints, preventing over 55% of predicted titer drop events and enabling data-driven intelligent fermentation control.

Process Engineer at Samsung Biologics *Sep 2024 - Feb 2025*

Validated Plant 5 medium preparation and media hold vessel systems, supporting operations up to 15,000 L scale, with a focus on CIP cycle development under GMP environment. Assisted in establishing cleaning parameters including flow rate, temperature, chemical concentration, and contact time to ensure effective residue removal and compliance with validation criteria.

Undergraduate Researcher at Champion Lab, Georgia Institute of Technology *Jan 2023 - May 2024*

- Anti-fungal nanotextured steel for medical implants

Conducting fabrication of nanotextured stainless steel by electroetching to reduce fungal adhesion through contact-killing. Performing confocal microscopy, scanning electron microscopy, flow cytometry, and kinetics assay to assess the efficacy of the contact-killing of the nanosteel on *Candida albicans* and *Fusarium oxysporum*.

R&D Summer Intern, Samsung Biologics

Jun 2023 - Aug 2023.

Undergraduate Researcher, Georgia Tech and Children's Hospital of Atlanta
Research Intern, SNU College of Medicine

Aug 2022 - May 2023
May 2022 - Jul 2022

Leadership

Founder of Proud to be First

Oct 2023 - May 2024

Established and managed an online central hub, serving over 500 first-generation students on campus while facilitating seamless access to vital resources, fostering a supportive community, and empowering them on their educational journey.

Advocacy Chair of American Medical Student Association (AMSA)

Aug 2022 - May 2023

Managed a diverse array of events, including networking nights and volunteer opportunities with the Atlanta Food Bank. Events resulted in a measurable 30% increase in community engagement and a record-breaking 20% rise in volunteer participation rates.

Employment History

Teaching Assistant, Department of Chemical and Biomolecular Engineering, KAIST

Jan 2026–Jun 2026

Course: CBE.20007 MyCBE Career design in chemical and biomolecular engineering
Designed and managed key course programs, including professor-student networking sessions and graduate student mentoring initiatives, to support undergraduate career development. Served as a technical mentor, sharing hands-on research experiences in AI-Bio convergence and advising undergraduate students on emerging interdisciplinary fields. Coordinated and led an experiential field trip to the Korea Research Institute of Bioscience and Biotechnology (KRIBB), providing students with direct exposure to national-level research facilities.

New Student Guide (NSG), KAIST

Sep 2025 – Dec 2025

Mentored and supported a dedicated cohort of 8 incoming international students to facilitate their campus transition and cultural adaptation to South Korea. Mitigated linguistic and cultural barriers by providing ongoing logistical and emotional support, contributing to a more inclusive and global campus environment. Maintained long-term support networks with students beyond the official program conclusion, ensuring continuous guidance and successful academic integration.

PLUS (Peer-Led Undergraduate Study) Program Assistant, Georgia Tech

Jan 2024 – May 2024

Oversaw the observation of peer leaders and completed performance assessments of 20+ peer educators. Assisted in planning and executing end of exam review hosting 1200+ students. Supervised new hire leaders through one-on-one meetings and feedback sessions. Increased total visits to supplemental instruction program by 30%.

PLUS Leader, Georgia Tech

Jan 2023 – Jan 2024

Facilitated weekly supplemental review sessions for Calculus, Chemistry, and Physics, guiding 30+ students per session through collaborative problem-solving. Fostered critical thinking by leading concept-based discussions that reinforced lecture material. Promoted effective study techniques and course resources to help students strengthen independent learning skills.

Samsung Biologics R&D Intern

Jun 2023 - Aug 2023

Spearheaded the platformization of forced degradation studies for API (monoclonal antibody and fusion peptide), resulting in a 30% reduction in turnaround time and a 20% increase in analytical efficiency. API initiative laid the foundation for a more streamlined CMC service model, facilitating quicker decision-making processes and enhancing overall analytical rigor. Utilized a diverse array of analytical techniques, including SEC-HPLC, IE, and ELISA assays.

Resident Assistant, Georgia Tech

Aug 2022 – Dec 2023

Fostered the development of relationships among 100+ residents. Planned and implemented programs to assist international and transfer students in transitioning to campus. Counseled and advised first year

students on academic and personal questions and concerns. Developed and conducted programs on diversity, chemical abuse, personal development, relationships, and academic performance. Managed administrative duties: budgeting, maintenance requests, incident reports, and room transfers.

Proscribe - Medical Scribe

Jan 2022 – Aug 2022

Documented medical history of over 2000 patients; records including CC, HPI, PMHx, and PSurgHx. Used Meditech Electronic Health Records system to record patients medical data, care plans and diagnosis. Monitored patient medical charts and recognized data discrepancies. Reported medical data discrepancies to physicians and nursing assistants. Assisted physicians in accessing all relevant test results and medical history documents.

Accomplishments and Conferences

Kim Young-Han Global Leader Scholarship (\$2700)	2026
Best Oral Award, 16th Chemical and Biomolecular Engineering Graduate Student Symposium	May 2026
Oral presentation at Korean Society for Biotechnology and Bioengineering	Apr 2026
<i>A Multimodal Model for Condition-Aware Gene Expression Prediction</i>	
Published <i>Artificial Intelligence for Multiscale Biological System Design and Optimization</i> for the Comprehensive Biotechnology, 4th Edition	Oct 2025
Oral presentation at Korean Society for Biotechnology and Bioengineering	Apr 2025
<i>Predictive Machine Learning Model for Optimizing 2,3-Butanediol Production through Microbial Fermentation</i>	
Poster presentation at the American Chemical Society	Mar 2024
<i>Nanotextured Stainless Steel for Antifungal Applications</i>	
Teams for Tech Fellowship	Oct 2023 – May 2024
Millennium Fellowship	Aug 2023 - Mar 2024
President's Undergraduate Research Travel Award	Nov 2023
Create-X Startup sponsored by Idea to Prototype (I2P) at Georgia Tech	Jan 2022 - May 2022
Honorable Mention: Aspirations in Computing (AiC), National Center for Women & Information Technology	May 2020